$\boxed{L30}$ 05/11/2020

$n_p :=$ # of open rhp poles of $P(s)$
$n_c :=$ # of open rhp poles of $\hat{K}(s)$

The system in fig (ii) is internally stable if and only if

(i) no. of open rhp poles of $P\hat{K} = n_c + n_p$

(ii) $(I - P\hat{K})^{-1}$ is stable.

Well-posedness is inherently assumed

Proof: $P = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array}\right]$ & $\hat{K} := \left[\begin{array}{c|c} \hat{A} & \hat{B} \\ \hline \hat{C} & \hat{D} \end{array}\right]$ $\qquad P\hat{K} = \left[\begin{array}{cc|c} A & B\hat{C} & B\hat{D} \\ 0 & \hat{A} & \hat{B} \\ \hline C & D\hat{C} & D\hat{D} \end{array}\right]$

space(K) poles of T.F. conv. A,B,C

$$(I - P\hat{K})^{-1} = \left[\begin{array}{c|c} \overbrace{\begin{bmatrix} A & B\hat{C} \\ 0 & \hat{A} \end{bmatrix} + \begin{bmatrix} B\hat{D} \\ \hat{B} \end{bmatrix}(I - D\hat{D})^{-1}\begin{bmatrix} C & D\hat{C} \end{bmatrix}}^{\bar{A}} & \overbrace{\begin{bmatrix} B\hat{D} \\ \hat{B} \end{bmatrix}(I - D\hat{D})^{-1}}^{\bar{B}} \\ \hline \underbrace{(I - D\hat{D})^{-1}\begin{bmatrix} C & D\hat{C} \end{bmatrix}}_{\bar{C}} & \underbrace{(I - D\hat{D})^{-1}}_{\bar{D}} \end{array}\right]$$

Only if: Given the system is internally stable. $\Rightarrow (I - P\hat{K})^{-1} \in RH_\infty \Rightarrow (I - P\hat{K})^{-1}$ is stable.

$\rightarrow$ * $\bar{A} = \tilde{A}$ $\qquad \boxed{(\bar{A}, \bar{B}) \text{ is stabilizable} \quad (\bar{C}, \bar{A}) \text{ is detectable}}$

* $\longrightarrow$ * $(\bar{A}, \bar{B}, \bar{C})$ is stabilizable and detectable $\Longleftrightarrow \left(\begin{bmatrix} A & B\hat{C} \\ 0 & \hat{A} \end{bmatrix}, \begin{bmatrix} B\hat{D} \\ \hat{B} \end{bmatrix}, \begin{bmatrix} C & D\hat{C} \end{bmatrix}\right)$ is stabilizable & detectable

You will do

$\Updownarrow$

Unstable eigenvalues of $\begin{bmatrix} A & B\hat{C} \\ 0 & \hat{A} \end{bmatrix}$ are controllable and observable.

$\Longleftrightarrow$ There is no pole-zero cancellation of the unstable eigenvalues of $\begin{bmatrix} A & B\hat{C} \\ 0 & \hat{A} \end{bmatrix}$

$\xrightarrow{(ii)}$

$\Longleftrightarrow$ No. of open rhp poles of $P\hat{K}$ must be $n_c + n_p$.

If: $\bar{A}$ is Hurwitz $\Leftrightarrow \tilde{A}$ is Hurwitz $\Leftrightarrow$ Internal stability. #

$\boxed{}$ $\bar{A} = \tilde{A}$

Proof:

$$\begin{cases} \bar{A} = \begin{bmatrix} A & B\hat{C} \\ 0 & \hat{A} \end{bmatrix} + \begin{bmatrix} B\hat{D} \\ \hat{B} \end{bmatrix} (I - D\hat{D})^{-1} \begin{bmatrix} C & D\hat{C} \end{bmatrix} \\ \tilde{A} = \begin{bmatrix} A & 0 \\ 0 & \hat{A} \end{bmatrix} + \begin{bmatrix} B & 0 \\ 0 & \hat{B} \end{bmatrix} \underbrace{\begin{bmatrix} I & -\hat{D} \\ -D & I \end{bmatrix}^{-1}} \begin{bmatrix} 0 & \hat{C} \\ C & 0 \end{bmatrix} \end{cases}$$

$\boxed{\text{Woodbury matrix identity}}$

$$\begin{bmatrix} I & -\hat{D} \\ -D & I \end{bmatrix}^{-1} = \begin{bmatrix} I + \hat{D}(I - D\hat{D})^{-1}D & \hat{D}(I - D\hat{D})^{-1} \\ (I - D\hat{D})^{-1}D & (I - D\hat{D})^{-1} \end{bmatrix}$$

Push-through identity

$(I + UV)^{-1}U = U(I + VU)^{-1}$.

$$\begin{bmatrix} A & 0 \\ 0 & \hat{A} \end{bmatrix} + \begin{bmatrix} B & 0 \\ 0 & \hat{B} \end{bmatrix} \begin{bmatrix} I & -\hat{D} \\ -D & I \end{bmatrix}^{-1} \begin{bmatrix} 0 & \hat{C} \\ C & 0 \end{bmatrix}$$

$$= \begin{bmatrix} A & 0 \\ 0 & \hat{A} \end{bmatrix} + \begin{bmatrix} B & 0 \\ 0 & \hat{B} \end{bmatrix} \begin{bmatrix} I + \hat{D}(I - D\hat{D})^{-1}D & \hat{D}(I - D\hat{D})^{-1} \\ (I - D\hat{D})^{-1}D & (I - D\hat{D})^{-1} \end{bmatrix} \begin{bmatrix} 0 & \hat{C} \\ C & 0 \end{bmatrix}$$

$$= \begin{bmatrix} A & 0 \\ 0 & \hat{A} \end{bmatrix} + \begin{bmatrix} B\hat{D}(I - D\hat{D})^{-1}C & B(I + \hat{D}(I - D\hat{D})^{-1}D)\hat{C} \\ \hat{B}(I - D\hat{D})^{-1}C & \hat{B}(I - D\hat{D})^{-1}D\hat{C} \end{bmatrix} = \bar{A}$$

e.g. $P = \begin{bmatrix} 1/s-1 & 0 \\ 0 & 1/s+1 \end{bmatrix} \sim \begin{bmatrix} 1/s^2-1 & 0 \\ 0 & 1 \end{bmatrix}$ $\hat{K} = \begin{bmatrix} \frac{-(s-1)}{s+1} & -1 \\ 0 & -1 \end{bmatrix} \sim \begin{bmatrix} 1/s+1 & 0 \\ 0 & s-1 \end{bmatrix}$

Internal stable?

$n_p = 1$ $n_c = 0$

\# rhp poles for $P\hat{K} = 1$

$(I - P\hat{K})^{-1} \leftarrow$ This is unstable

Closed loop system is unstable.

G^{-1} is stable $\Rightarrow \det(G) \leftarrow$ zeros are $\mathbb{C}^-$

Suppose $\hat{K} \in \mathcal{RH}_\infty$. Then system in fig (11) is internally stable if and only if it is well-posed & $P(I-\hat{K}P)^{-1} \in \mathcal{RH}_\infty \leftarrow (I-P\hat{K})^{-1}P \in \mathcal{RH}_\infty$

Proof: Only if: System is internally stable,

$$\begin{bmatrix} I + \hat{K}(I-P\hat{K})^{-1}P & \hat{K}(I-P\hat{K})^{-1} \\ (I-P\hat{K})^{-1}P & (I-P\hat{K})^{-1} \end{bmatrix} \in \mathcal{RH}_\infty$$

$$\Rightarrow (I-P\hat{K})^{-1}P \in \mathcal{RH}_\infty \Rightarrow P(I-\hat{K}P)^{-1} \in \mathcal{RH}_\infty$$

If: $P(I-\hat{K}P)^{-1} \in \mathcal{RH}_\infty \Rightarrow (I-P\hat{K})^{-1}P \in \mathcal{RH}_\infty$

$$(I-P\hat{K})^{-1} = I + P(I-\hat{K}P)^{-1}\hat{K}$$

$$= I + \underbrace{(I-P\hat{K})^{-1}P}_{\mathcal{RH}_\infty}\underbrace{\hat{K}}_{\mathcal{RH}_\infty} \Big\} \in \mathcal{RH}_\infty$$

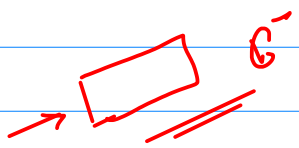
$$\therefore \begin{bmatrix} I & -\hat{K} \\ -P & I \end{bmatrix} \in \mathcal{RH}_\infty$$

* If $Q, M \in \mathcal{BL}(U, V)$, then $(I_U - QM)^{-1} = I_U + Q(I_V - MQ)^{-1}M$

Let $P \in \mathcal{RH}_\infty$. Then system is internally stable iff $\hat{K}(I-P\hat{K})^{-1} \in \mathcal{RH}_\infty$

$P, \hat{K} \in \mathcal{RH}_\infty$. Then system is internally stable iff $(I-P\hat{K})^{-1} \in \mathcal{RH}_\infty$

$P = \left(\frac{s-1}{s+2}\right) \in \mathcal{RH}_\infty \quad \hat{K} = \left(\frac{1}{s-1}\right) \notin \mathcal{RH}_\infty \quad \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} \to \begin{pmatrix} e_1 \\ e_2 \end{pmatrix} \quad \begin{bmatrix} I & -\hat{K} \\ -P & I \end{bmatrix}^{-1}$

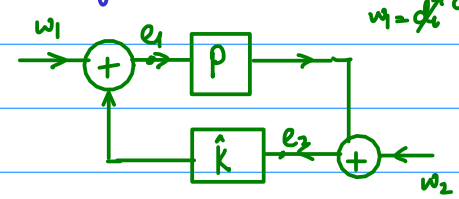


$$\begin{bmatrix} e_1 \\ e_2 \end{bmatrix} = \begin{bmatrix} \frac{s+2}{s+1} & \boxed{\frac{s+2}{s^2-1}} \\ \frac{s-1}{s+1} & \frac{s+2}{s+1} \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}$$

$\notin \mathcal{RH}_\infty$

Not internally stable

$w_2 = -r + \cancel{y_p}^{\,0}$, $w_1 = \cancel{d_i}^{\,0}$



$P\hat{K}$ $\quad \hat{K}P$

$w_2 \to e_1 \to \boxed{y_p} \quad \left(\frac{s+2}{s^2-1}\right) \times \left(\frac{s-1}{s+2}\right) = \boxed{\frac{1}{s+1}}$

Cancellation of unstable pole/zero

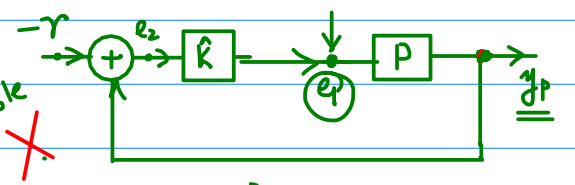
$w_2 \to e_1 \quad \frac{s+2}{s^2-1}$

$\boxed{P \in \mathcal{RH}_\infty}$

$K \in \mathcal{RH}_\infty$

$P, K \in \mathcal{RH}_\infty$

$s+1 \quad -1$



Stable ✗

$1 - \hat{K}P = 1 - \left(\frac{1}{s-1}\right)\left(\frac{s-1}{s+2}\right) = \frac{s+1}{s+2}$

$\frac{y_p}{r} = \frac{-\hat{K}P}{\boxed{1-\hat{K}P}}$